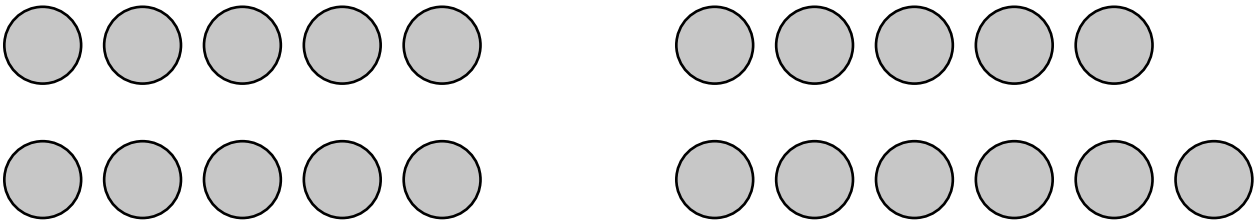
Two players take turns. Whoever takes the last counter wins. On a track, the token starts on the number of counters and moves toward 0 by the number of counters taken, and whoever puts it on 0 wins.

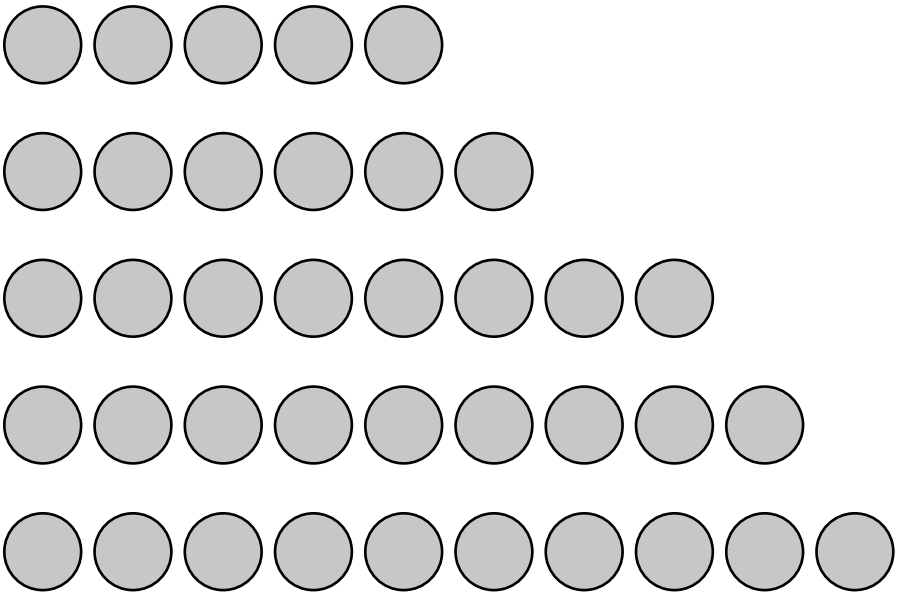
Problem 6: With moves of 1, 3 or 4, would you rather go first or second with a pile of 50? With a pile of 100? If you would go first, write how many counters you would take. Explain how you know.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11
	21	22	23	24	25	26	27	28	29	30

Problem 7: Now make two piles. On each turn, take 1, 3 or 4 counters from one of the piles. Whoever takes the last counter on the table wins. With piles of 5 and 5, and with piles of 5 and 6, would you rather go first or second? Explain why your way of playing wins whatever your opponent does.



Problem 8: Take 1 or 4 counters on each turn. Play each pile with your partner until you are sure whether you would rather go first or second, and circle your choice. If you would rather go first, also circle the counters you take on your first turn.



first second

first second

first second

first second

first second

Problem 9: For each rule, colour every square where you would like to leave the token for your opponent.

Take 1 or 4 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11

Take 1, 3 or 5 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11